

Toward Task-Aware Reuse of Cellular Molecular Receivers: History-Aware Prediction on Experimental FGF2–ERK Data

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Abstract—We ask which observable information predicts a cellular molecular receiver’s response to repeated chemical commands and whether it can shorten the wait between commands. On public experimental FGF2–ERK single-cell trajectories, a nested post hoc comparison with a fixed, previously calibrated mechanistic reference (the authors’ B3 model) showed that, on a protocol unseen in training, reporter filtering reduced the 10-min-ahead error of a current-state predictor by 9.1%, the B3 increment reduced the error of the filtered predictor by 4.5% and that of a lagged non-mechanistic predictor by 6.9%, and explicit input-history features added only 1.0%; the mechanism raised the error before an imminent command. In simulation on B3, rules that wait until an estimated success probability reaches a calibrated threshold completed a fixed probe task 4.8 min faster than fixed waits that already use the known previous command, at similar success and in all 20 calibration/evaluation splits. A smoothed reporter matched the history-aware estimator at the reference noise, the latter helped only at higher noise, and both lost reliability under correlated noise. No result concerns measured task reliability.

Index Terms—Molecular communication, cellular receivers, mechanism-informed prediction, receiver reuse, waiting rules, FGF2, ERK.

I. INTRODUCTION

Molecular communication (MC) with living receivers asks a chemical command to produce a specified response in a cell. Reception then means executing the requested response on time, and a previous command can change the next response through delivery persistence, receptor dynamics, intracellular feedback, or cellular state. For repeated tasks the operational question is when the receiver can be used again at matched reliability. We ground this question in recorded biology: FGF2 stimulation of PC-12 cells with ERK activity read by a FRET reporter [1], treated as a continuous receiver output without retroactive symbols or cell-fate labels.

Two lines of work bound the claim. In cell dynamics, the FGF2 data were used to infer signaling architecture, including feedback and a nontrivial washout response [1]; stimulation frequency shapes ERK responses [2]; automated chemical feedback has controlled mammalian signaling [3]; and a learned dynamical model enabled model predictive control of thousands of single cells [4]. In MC, memory is classical: diffusion and ligand binding produce intersymbol interference and reception noise [5]–[7], which guard intervals, ISI-aware modulation and adaptive threshold detection mitigate [8],

[9]. Semantic systems learn task-relevant representations [10], [11], interpretable reaction–diffusion learning has been proposed [12], experimental yeast pheromone links characterize biological receiver noise [13], genetic-circuit receivers are designed computationally [14], calcium signaling with refractory behavior has been controlled [15], error control has been adapted to cellular noise [16], and long-term experimental MC exposes reuse tradeoffs [17]. The open question is which information makes a *next task* predictable for a living receiver, whose memory lies in its biochemistry rather than only in the channel, and how much a waiting decision gains from it.

This paper makes two bounded contributions. (i) On experimental FGF2–ERK trajectories, a nested comparison quantifies in which forecast phases reporter filtering, a fixed calibrated mechanistic increment and explicit input-history features add predictive value, against a lagged non-mechanistic baseline and with leave-one-protocol-out (LOPO) validation in which predictor tuning is restricted to the training protocols. (ii) In simulation on B3, for a fixed probe task, we quantify how much feedback and history estimation add to fixed waits that already use the known previous command, under the same calibration constraint, with no-probe, noise, bandwidth and mismatch checks. The two parts are separate evidence streams (Fig. 1): the experimental predictors do not enter the simulated waiting rule.

II. SYSTEM AND TASK MODEL

Let $u(t) \geq 0$ denote the commanded FGF2 concentration in ng/ml. Delivery to a receiver is represented abstractly as

$$\begin{aligned} c_R(t) &= \mathcal{T}_\theta[u](t), & \dot{x}_j &= f_\theta(x_j, c_R), \\ z_j &= g(x_j), & y_{j,i} &= z_j(t_i) + \nu_{j,i}, \end{aligned} \quad (1)$$

where c_R is the local concentration, x_j the unobserved state of cell j , z_j its noise-free normalized reporter, and $y_{j,i}$ the observation with noise $\nu_{j,i}$ at frame t_i . The operator $\mathcal{T}_\theta$ can include delivery lag and washout; neither it nor c_R is calibrated here, and the FRET ratio is not a molecular count or a cell-fate measurement. The available information $\mathcal{H}_t$ contains the executed commands and the observations whose feedback has arrived by t ; offline, each observation is assumed available at its recorded time.

We fix the amplitude and duration of the next command and decide only when to send it. After a previous command

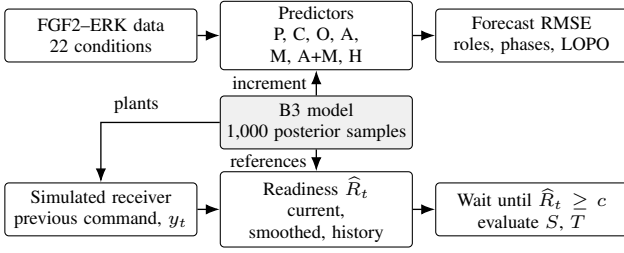


Fig. 1. Two evidence streams. Top: experimental trajectories train and test the predictors, with B3 as a fixed, previously calibrated mechanistic increment. Bottom: B3 posterior samples act as simulated receivers and as estimator references for the waiting rule (Sec. V-C). The experimental predictors do not enter the waiting rule; delivery and latent states are not calibrated.

ending at time 0, a 5-min probe at the same amplitude is sent at $\tau \in \mathcal{T} = \{0, 2, \dots, 120\}$ min. With z_j^τ the noise-free response when the probe is sent at τ after the recorded history, and $r_{\text{naive},j}$ the same receiver's rise to the probe without a previous command, the task outcome and readiness are

$$S_j(\tau) = \mathbf{1}_{\left\{ \max_{s \in [\tau, \tau+W]} z_j^\tau(s) - z_j^\tau(\tau) \geq \kappa r_{\text{naive},j} \right\}}, \quad (2)$$

$$R_{j,t} = \mathbb{P}(S_j(t) = 1 \mid \mathcal{H}_t),$$

with task window $W = 20$ min and $\kappa = 0.5$, a prescribed probe criterion rather than one derived from a biological task (W and the penalty D below are likewise prescribed assessment settings). Observation noise affects the feedback used for decision-making; success is scored from the noise-free model response, and decisions use only observations available by τ . The counterfactual $r_{\text{naive},j}$ is a simulation benchmark; an experiment would have to measure or estimate it, or replace it by a fixed measurable threshold. The failure-penalized completion time is $T = \tau + W$ (the end of the assessment window, not the time the criterion is first met) on success and $D = 140$ min on failure; the design objective is to minimize $\mathbb{E}[T]$ subject to $\mathbb{P}(S = 1) \geq q = 0.9$, an empirical target rather than a coverage guarantee. The experimental data have no task labels, so Eq. (2) is evaluated only in simulation.

III. MECHANISM-INFORMED PREDICTION AND WAITING

A. Nested Predictors

Separate predictors are fit for horizons $h = 2$ and 10 min after a fixed 10-min observed warmup. Every learned predictor receives the same known future command (its exposure in 2-min bins, the concentration at $t + h$, and h); future reporters are targets only. The current-state set C contains $u_t, u_t^2, y_t, y_t^2, u_t y_t$, the 2-min slope $(y_t - y_{t-2})/2$ (so C is not memoryless) and the future-command features. Causal filters with scales $\tau \in \{2, 10, 60\}$ min summarize input and reporter history,

$$q_\tau(t) = \frac{1}{\tau} \int_0^t e^{-(t-s)/\tau} u(s) ds, \quad (3)$$

$$r_{\tau,i} = e^{-\Delta_i/\tau} r_{\tau,i-1} + (1 - e^{-\Delta_i/\tau}) y_i,$$

with $r_{\tau,0} = y_0$ and $\Delta_i = t_i - t_{i-1}$; the scales are computational choices, not identified rate constants. O adds the reporter filters

r_τ to C . As a non-mechanistic history baseline, A adds to C the reporter and command values of the previous $L \in \{5, 15, 30\}$ frames (10–60 min). As a mechanism we execute the authors' B3 model with their 1,000 posterior parameter samples. With $f_s(t)$ the noise-free reporter of sample s under the recorded command, the samples are weighted by each cell's past reporter,

$$w_s(t) \propto \exp \left[- \sum_{t_i \leq t} (y_i - f_s(t_i))^2 / (2\sigma^2) \right], \quad (4)$$

$$\hat{\delta}_{t,h} = \sum_s w_s(t) [f_s(t+h) - f_s(t)],$$

a fixed-parameter particle approximation of the B3 belief that uses the same information as the learned predictors. M adds $\hat{\delta}_{t,h}$ to O , $A+M$ adds it to A , and H adds the input filters q_τ and the cumulative exposure $\int_0^t u(s) ds$ to M . The B3 increment already uses the command history and past observations, so H tests only the additional value of explicit input-history features. Persistence P ($\hat{y}_{t+h} = y_t$) needs no training. Each set is fit by ridge regression with equal weight per condition and standardization from training rows; the penalty (from 10^{-8} to 10^4), σ (seven values) and L are selected on validation. The B3 samples express population-mean uncertainty, not cell heterogeneity.

B. Calibrated Waiting Rules

A waiting rule probes at

$$\tau = \begin{cases} \inf\{t \in \mathcal{T} : \hat{R}_t \geq c\}, & \text{if such a } t \text{ exists,} \\ \max \mathcal{T}, & \text{otherwise.} \end{cases} \quad (5)$$

Three estimators of R_t are compared. The current estimator uses only y_t , through a Gaussian kernel over the (reporter, success) states of reference receivers; the smoothed estimator applies the same kernel to a causal exponential smoothing of the observations (scale 2–16 min); the history-aware estimator weights reference receivers under the known command history by all observations so far, as in Eq. (4), with likelihood s.d. equal to the noise s.d. floored at 0.001 (a floor of 0.005 is a pre-specified variant). Reference receivers are the calibration samples (leaving out the receiver itself during calibration); evaluation samples never enter references, thresholds or waits. Because per-frame calibration does not guarantee the reliability of an adaptive stopping rule, the whole rule is calibrated: for the current and history estimators we take the first threshold on the grid $\{0.5, \dots, 0.99\}$ whose calibration success reaches q , a heuristic guided by the completion-time objective; for each smoothing scale we take a feasible threshold the same way and then the scale with the smallest calibration mean completion; if nothing is feasible, the largest threshold is used and the rule is reported as missing the target. As the adaptive rules know the previous command, the main fixed comparator also uses it: two waits (τ_3, τ_{30}) are chosen jointly over the 61^2 grid pairs to minimize the equal-weight calibration mean completion subject to pooled calibration success $\geq q$; a variant requires q for each history, and a single wait for both histories is also reported. Choosing the fastest feasible threshold instead of the first and a finer threshold grid are sensitivity analyses.

TABLE I
PROTOCOLS, ROLES AND SINGLE-CELL TRAJECTORIES (22 CONDITIONS,
1,565 CELLS, 2-MIN FRAMES). B3 USE: CONCENTRATIONS WHOSE MEANS
THE AUTHORS FITTED (FIT) OR USED IN THEIR ARCHITECTURE
COMPARISON (ARCH.).

Protocol	Role	ng/ml	Cells	B3 use
Sustained	Train	0.25, 2.5, 25, 250	248	fit 2.5, 250
3/20 pulses	Train	2.5, 25, 250	207	fit 2.5, 250
Single 5 min	Val.	0.25, 2.5, 25, 250	370	arch. 2.5, 250
Single 10 min	Test A	0.25, 2.5, 25, 250	248	–
Single 60 min	Test B	2.5, 25, 250	167	–
Mixed	Test C	0.25, 25	171	–
Mixed	Viewed	2.5, 250	154	arch. 2.5, 250

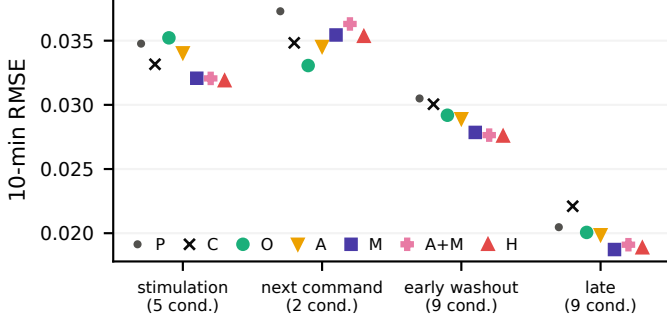


Fig. 2. Condition-equal 10-min RMSE on Tests A–C by forecast phase for all predictors (axis not starting at zero).

IV. DATA AND EVALUATION PROTOCOL

A. Data and Roles

All numerical files come from the paper-linked `Mijan/LFNS_MSB` repository at a pinned commit with verified file hashes [18], [19]. Commands follow the authors’ model XML with half-open pulses; mixed pulses span [1, 4), [24, 54) and [114, 119) min. Model time 0 is the stimulation onset, and per-cell normalization uses only earlier frames (a mean over raw 2–38 min in the exports and a median over raw 10–30 min, reconstructed by us, for the mixed protocol whose origin is raw 42 min), so every forecast uses information available at its origin. Table I lists the 22 conditions. The roles were frozen before any newly used response file was downloaded: Test A is a new protocol, Test B’s [0, 60) timing was inferred from the authors’ B3 simulations (± 1 min sensitivity), and Test C uses new concentrations recorded in the session of the viewed mixed blocks, on which an earlier pilot was tested. Every analysis below was frozen in its own protocol before running, but all are post hoc with respect to inspected data.

B. Mechanism, Validation and Scores

B3 is executed by parsing the authors’ model files; no equation is retyped, no parameter is refit, and our posterior-predictive mean matches their exported predictions within 8.5×10^{-6} . Because the authors fitted B3 to the sustained and 3/20 means and compared architectures on the single 5-min and viewed mixed blocks at 2.5 and 250 ng/ml (Table I),

M, A+M and H use a calibrated reference, not a model blind to those protocols. In LOPO each of six protocol groups is held out in turn, with predictor tuning restricted to the training protocols (σ on the training groups; penalty and L by inner leave-one-group-out) while B3 remains fixed; only the 10- and 60-min pulse protocols were absent from the original B3 fitting and architecture comparison. Forecast rows take the first matching phase in the order stimulation (command on at t), next command (a pulse starts in $(t, t+h]$), early washout (last pulse ended at most 30 min before t), and late; after its warmup Test A contains only washout rows. The score is the square root of the equally weighted condition MSE,

$$\text{RMSE}_{\text{cond}}(h) = \left[\frac{1}{|D|} \sum_{d \in D} \frac{1}{N_{d,h}} \sum_{i \in d} (\hat{y}_{i+h} - y_{i+h})^2 \right]^{1/2}, \quad (6)$$

reported with per-condition directions. Rows and cells are not independent interventions and experimental-day identifiers are unavailable, so no significance test is reported.

V. RESULTS AND DISCUSSION

A. Nested Prediction

Table II(a) reports the nested predictors. On Test A at 10 min, reporter filtering reduced the error by 9.1% relative to C, the B3 increment by 4.5% relative to O, and explicit input-history features by 1.0% relative to M, giving 0.02098, 8.2% below persistence; these reductions use different baselines and are not additive. The H-versus-M gain was small and not consistent across conditions (two of four on Test A; 0.4% and 0.1% on B and C). The lagged baseline A was 2.3% worse than O on Test A (better on B and C), but adding the B3 increment to it still reduced the error by 6.9% on Test A (four of four concentrations), 3.6% on B and 4.2% on C, and H and A+M were within 1.4%. The mechanistic increment thus adds value beyond a directly represented input and reporter history, whereas explicit input history adds little once the mechanism is present.

B. Forecast Phases and LOPO

By phase (Fig. 2), the B3 increment reduced the error of O in every phase except the next-command window (by 8.9% during stimulation, 4.6% in early washout and 6.6% late), and that of A in the same phases. During stimulation the reporter filters hurt (O 0.03522 versus C 0.03316). The next-command window is covered by two mixed conditions with ten forecast origins each: the increment helped at 0.25 ng/ml (O 0.03652, M 0.03485; A 0.03774, A+M 0.03601) but raised the error at 25 ng/ml (O 0.02920, M 0.03603; A 0.03102, A+M 0.03659), so the pooled error rose from 0.03306 to 0.03544. This limits the mechanism for predicting responses to an imminent command despite its benefits during washout. In the corrected LOPO (Table II(b)), H beat M in all six folds (by 0.1–2.0%) and A+M beat A in all six (by 1.6–13.7%), including both protocols unseen by B3; with the 10-min pulse held out, however, O was best and M was 1.1% worse than O. The sustained and 3/20 folds contain conditions used to fit

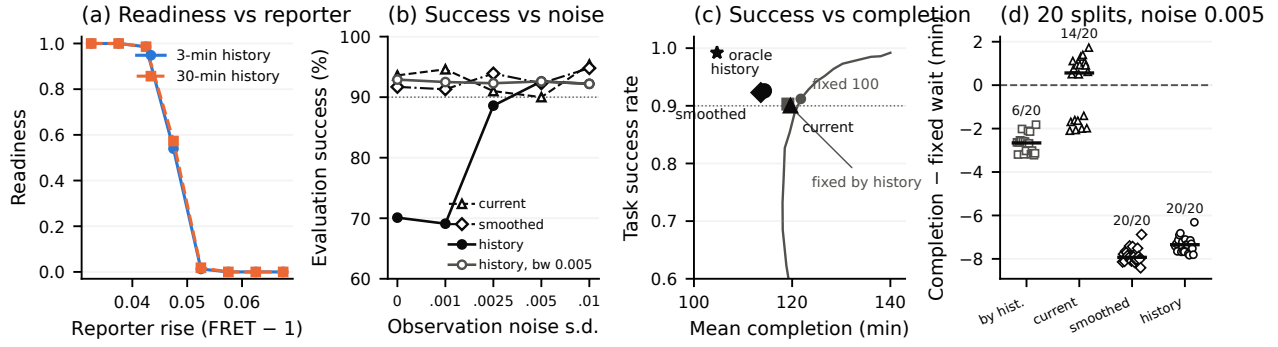


Fig. 3. Calibrated waiting on the authors' B3 model (25 ng/ml; task: probe rise $\geq 50\%$ of the naive rise within 20 min). (a) Readiness versus the current noise-free reporter after 3- or 30-min commands. (b) Evaluation success versus observation noise for the current, smoothed and history rules (likelihood s.d. floored at 0.001) and the history rule with floor 0.005; dotted: target 0.9. (c) Success versus mean failure-penalized completion at noise 0.005 on the primary split; gray curve: single fixed waits; square: history-conditioned fixed waits. (d) Twenty calibration/evaluation splits with fresh noise: completion minus that of the single fixed wait (bars: medians; numbers: splits whose evaluation success reached 0.9). Thresholds and waits were chosen on calibration samples. Simulation of one model structure, not measured cells.

TABLE II

CONDITION-EQUAL 10-MIN RMSE UNLESS NOTED. (A) VAL.: SINGLE 5-MIN PULSE; A: SINGLE 10-MIN PULSE; B: SINGLE 60-MIN PULSE (INFERRED TIMING); C: MIXED PULSES AT 0.25 AND 25 NG/ML. P AND C ARE PRE-REGISTERED; O, A, M, A+M AND H ARE POST HOC. BOTTOM ROWS COUNT CONDITIONS WITH LOWER MSE. (B) CORRECTED LOPO WITH PREDICTOR TUNING RESTRICTED TO THE TRAINING PROTOCOLS; B3 PRIOR USE FROM THE AUTHORS' INFERENCE CONFIGURATION. BOLD: SMALLEST PER COLUMN (A) OR ROW (B).

(a) Fixed roles					
Model	Val.	A	B	C	A, 2 min
P: persistence	0.02353	0.02287	0.02651	0.02876	0.01302
C: current + slope	0.02503	0.02442	0.02715	0.02825	0.01342
O: C + reporter filters	0.02403	0.02220	0.02715	0.02827	0.01283
A: C + lags	0.02343	0.02272	0.02565	0.02769	0.01300
M: O + B3 increment	0.02197	0.02120	0.02448	0.02658	0.01263
A+M: A + B3 increment	0.02161	0.02114	0.02474	0.02652	0.01270
H: M + input history	0.02195	0.02098	0.02439	0.02656	0.01260
H beats M	1/4	2/4	2/3	1/2	2/4
M beats O	4/4	3/4	2/3	2/2	3/4
A+M beats A	4/4	4/4	3/3	1/2	4/4
O beats C	4/4	4/4	2/3	1/2	4/4
(b) Leave-one-protocol-out					
Held-out	O	M	H	A	A+M B3 prior use
Sustained	0.0210	0.0197	0.0194	0.0202	0.0193 fit 2.5, 250
3/20 pulses	0.0368	0.0289	0.0283	0.0329	0.0284 fit 2.5, 250
Single 5 min	0.0224	0.0215	0.0214	0.0220	0.0213 arch. 2.5, 250
Single 10 min	0.0208	0.0211	0.0210	0.0212	0.0208 none
Single 60 min	0.0270	0.0241	0.0237	0.0249	0.0242 none
Mixed, 0.25/25	0.0271	0.0262	0.0261	0.0271	0.0263 other conc. arch.

B3, so their mechanistic gains are not out-of-sample for the mechanism.

C. Calibrated Waiting on B3

Readiness and waiting were simulated on B3 with 25 ng/ml commands, a 3- or 30-min previous command and observations every 2 min with noise s.d. 0.005. Readiness depended almost only on the current reporter (Fig. 3(a)), and the registered candidate waits of 6–48 min never succeeded. Without

TABLE III

CALIBRATED WAITING ON THE PRIMARY SPLIT BY PREVIOUS COMMAND (NOISE 0.005): EVALUATION SUCCESS S AND MEAN FAILURE-PENALIZED COMPLETION T AFTER THE 3- AND 30-MIN COMMANDS. "FIXED 92/102": HISTORY-CONDITIONED FIXED WAITS (POOLED CONSTRAINT). POOLED VALUES IN THE TEXT AVERAGE THE TWO HISTORIES.

Rule	S_3 (%)	T_3 (min)	S_{30} (%)	T_{30} (min)
Current obs.	88.8	115.3	91.2	124.0
Smoothed (16 min)	92.6	108.6	92.0	118.6
History-aware	91.0	109.1	94.2	119.6
Fixed 100 min	97.4	120.5	85.0	123.0
Fixed 92/102 min	92.4	114.1	88.2	124.1
Fixed 90/106 min (each ≥ 0.9)	90.2	112.9	91.6	127.2
Oracle	100.0	99.0	98.4	110.4

a probe no state met the criterion (the largest no-probe rise was 47% of the naive rise); a causal criterion based on the probe-induced increment agreed with Eq. (2) for 97.0% of states and only added successes, and rescoring the rules with it raised every success rate without changing their order.

On the primary split (500 calibration and 500 evaluation samples), the history-aware rule reached 92.6% success at 114.3 min and the smoothed reporter 92.3% at 113.6 min (Fig. 3(c)). A single fixed wait of 100 min reached 91.2% at 121.8 min, but only 85.0% after the 30-min command (Table III). History-conditioned fixed waits of 92 and 102 min reached 90.3% at 119.1 min, so knowing the previous command explains part of the advantage: the history rule remained 4.8 min faster (paired 95% bootstrap interval 4.0–5.6 min) and the smoothed rule 5.5 min faster (4.7–6.2), against 7.4 min relative to the single wait; requiring 0.9 for each history gave 90.9% at 120.1 min. Over 20 splits with fresh noise (Fig. 3(d)), the history rule beat the history-conditioned waits in every split (by 4.0–5.6 min), which reached 0.9 evaluation success in only six splits, and the smoothed rule beat the history rule in every split (by 0.2–1.0 min). Choosing the fastest feasible threshold gave the same choices as the heuristic, and on a fine threshold grid smoothing stayed 0.2–0.9 min faster at matched

TABLE IV

SENSITIVITY AND MISMATCH OF CALIBRATED WAITING ON THE PRIMARY SPLIT. ENTRIES: EVALUATION SUCCESS (%) / MEAN FAILURE-PENALIZED COMPLETION (MIN); WAIT: SINGLE FIXED WAIT (MIN); BY HISTORY: HISTORY-CONDITIONED FIXED WAITS; *: CALIBRATION TARGET NOT REACHED (FALLBACK). ΔT_{H-S} : HISTORY MINUS SMOOTHED COMPLETION (MIN) WITH PAIRED 95% BOOTSTRAP INTERVAL. $\dagger$: FROZEN $\kappa = 0.5$ RULES EVALUATED UNDER AR(1) OBSERVATION NOISE ($\rho = 0.8$, S.D. 0.005) OR A REPORTER OFFSET OF $+2.5 \times 10^{-3}$.

κ	Noise	Wait	Fixed	By history	History	Smoothed	Current	Oracle	ΔT_{H-S}
0.4	0.005	80	92.2 / 103.1	90.0 / 100.2	91.6 / 96.2	93.7 / 95.5	93.9 / 103.1	100.0 / 85.1	+0.7 [−0.1, +1.4]
0.4	0.010	80	92.2 / 103.1	90.0 / 100.2	93.2 / 96.8	95.8 / 98.4	95.2 / 113.9	100.0 / 85.1	−1.6 [−2.3, −0.8]
0.5	0.005	100	91.2 / 121.8	90.3 / 119.1	92.6 / 114.3	92.3 / 113.6	90.0 / 119.7	99.2 / 104.7	+0.7 [+0.3, +1.1]
0.5	0.010	100	91.2 / 121.8	90.3 / 119.1	92.2 / 114.9	94.8 / 117.5	95.4 / 131.0	99.2 / 104.7	−2.6 [−3.0, −2.1]
0.6	0.005	120	81.5 / 140.0*	81.5 / 140.0*	80.7 / 134.2*	81.5 / 136.4*	81.5 / 139.6*	81.5 / 125.7	−2.2 [−2.5, −2.0]
0.6	0.010	120	81.5 / 140.0*	81.5 / 140.0*	81.4 / 137.0*	81.5 / 140.0*	81.5 / 140.0*	81.5 / 125.7	−2.9 [−3.3, −2.6]
0.5	AR(1) $\dagger$	100	91.2 / 121.8	90.3 / 119.1	82.0 / 117.3	84.3 / 116.9	93.7 / 123.5	99.2 / 104.7	+0.4
0.5	offset $\dagger$	100	91.2 / 121.8	90.3 / 119.1	97.5 / 117.6	98.8 / 117.9	96.0 / 123.2	99.2 / 104.7	−0.3

evaluation success.

The history estimator depends on its likelihood bandwidth (Fig. 3(b)): with the bandwidth tied to the noise it missed the calibration target at noise s.d. up to 2.5×10^{-3} (69.1–88.6% evaluation success) as its weights collapsed (median effective sample size 1.1 at zero noise versus 28.9 at 0.005), whereas the 0.005 floor reached the target at every level (92.3–92.9%). At doubled noise the current rule became slower than the fixed wait, and the history rule was 1.3–2.0 min faster than smoothing at matched success (Table IV). Under correlated noise that the estimators do not model, the frozen rules lost reliability (smoothed 84.3%, history 82.0%) while staying faster than fixed waiting; a reporter offset made them conservative (97.5–98.8% at about 118 min). At $\kappa = 0.4$ the history rule was 3.4–4.0 min faster than history-conditioned waits, and at $\kappa = 0.6$ even the oracle succeeded for only 81.5% of samples. In this B3 setting feedback saved about 5 min beyond knowing the previous command, smoothing sufficed at the reference noise, and model-based filtering paid off only at higher noise and needs a valid noise model; the results do not establish a waiting benefit from latent memory.

D. Implications for MC Design

In MC terms, a previous command acts on the next one like intersymbol interference whose memory sits in the receiver’s biochemistry rather than only in the channel [6], [8]. A fixed wait is a guard interval: conditioning it on the known previous command shortens it (92 and 102 instead of 100 min) but cannot equalize per-history reliability, whereas a feedback-driven wait adapts to the observed receiver state, much as adaptive threshold detection adapts to residual interference [9]. As reception noise does for molecular receivers [7], the observation-noise model decides which estimator such feedback needs, and a misspecified noise model breaks its calibration. A prospective test should randomize 3- versus 30-min preconditioning per chamber, extend probe waits to 60–120 min, measure the naive response in unpreconditioned controls of the same session, and count independent sessions rather than cells.

E. Evidence Boundaries

Three groups of limits remain. Data: the prediction comparisons are post hoc on inspected roles; in the frozen analysis of the main fresh test, history features alone improved on persistence by only 0.7%; Test C shares a session with inspected data and no experimental-day identifiers exist, so protocol generalization is not cross-day generalization; B3 had seen some protocols. Model: the waiting results come from one simulated model structure whose posterior samples express parameter uncertainty, not validated cell heterogeneity; the 20 splits reuse one finite posterior set rather than independent experiments; the 90% target has no coverage guarantee; the zero-noise failure of the history estimator reflects finite references and its bandwidth, not biology. Task: $\kappa = 0.5$ is a prescribed probe criterion adopted after an 80% version proved infeasible, the naive response is a simulation counterfactual, amplitude and duration were fixed, and the experimental predictors were not tested as decision models, so forecast gains do not imply better waiting decisions.

VI. CONCLUSION

On experimental FGF2–ERK trajectories, reporter filtering and a fixed calibrated mechanistic increment account for most of the 10-min prediction gain, also beyond a lagged non-mechanistic baseline; explicit input history adds little, and the mechanism is unreliable before an imminent command. In B3 simulations, feedback-driven waiting saved about 5 min beyond fixed waits that use the known previous command; smoothing sufficed at the reference noise, while model-based filtering helped only at higher noise and needs a valid noise model.

DATA AND CODE AVAILABILITY

Code, frozen results, protocols with hash records, and verification scripts are available at <https://github.com/ruifeng105/task-aware-cellular-mc>; one command re-runs the analysis and checks the reported values against a frozen reference. Sources are credited in [1], [18], [19]; author files keep their attribution and receive no new license.

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